

Curriculum Vitae
Brandon Pries, B.S.
Graduate Teaching Assistant, School of Physics, Georgia Institute of Technology
October 6, 2026

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Brandon Pries

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Education

Georgia Institute of Technology (Georgia Tech), Atlanta, GA

August 22, 2022 – Present

GPA 3.83/4.0; Major GPA 3.8/4.0

Doctor of Philosophy, Physics, College of Sciences

Expected Spring 2028

- Galaxies, cosmology, high-energy astrophysics
- Classical/quantum/statistical mechanics, electromagnetism, computational physics

Doctoral Minor in Higher Education

- Learner-centered teaching, metacognition, student motivation/engagement, active learning, backwards design, universal design for learning (UDL)

Dissertation Topic: *Formation and Evolution of Direct Collapse Black Holes as Supermassive Black Hole Seeds*

- Advisor: John Wise
- Thesis Committee: David Ballantyne, Tamara Bogdanović, Surabhi Sachdev

Michigan State University (MSU), East Lansing, MI

August 29, 2018 – May 7, 2022

GPA 3.965/4.0; Major GPA 4.0/4.0

Honors College

Bachelor of Science, Astrophysics, College of Natural Science (with High Honors)

May 7, 2022

- Exoplanets, telescopes, stars, galaxies
- Thermodynamics, classical/quantum/statistical mechanics, electromagnetism, computational physics

Minor in Mathematics

- Calculus, linear/abstract algebra, number theory, analysis, ordinary/partial differential equations

Minor in Computational Mathematics, Science, and Engineering (CMSE)

- Computational modeling, parallelization, high-performance computing (HPC), GPU acceleration, Python, C++, Bash, Linux, SLURM, Git/GitHub

Minor in Data Science

- Probability/statistics, hypothesis testing, sampling, model development/fitting/selection, R, SQL

Undergraduate Thesis: *Indirect Search for Dark Matter via Neutrinos from WIMP Annihilation with IceCube* (May 4, 2022)

- Advisors/Mentors: Tyce “Ty” DeYoung, Mehr U. Nisa

Professional Appointments

Graduate Teaching Assistant (GTA), Georgia Tech	August 2022 – May 2024; August 2025 – December 2025; August 2026 – Present
Graduate Research Assistant (GRA), Georgia Tech	May 2024 – August 2025; January 2026 – August 2026
Research Assistant, MSU	May 2022 – Present
Undergraduate Learning Assistant (ULA), MSU	January 2021 – May 2021; August 2021 – May 2022
Undergraduate Research Assistant, MSU	May 2019 – August 2019; May 2020 – May 2022
Professorial Assistant (PA), Honors College, MSU	August 2018 – May 2019; August 2019 – May 2020

Honors, Scholarships, and Awards

Tech to Teaching Certificate	March 10, 2026
CIRTL Associate Certificate	March 10, 2025
• Center for the Integration of Research, Teaching, and Learning	
Center for Relativistic Astrophysics (CRA) Travel Award	November 2024
• \$1,000 award for travel to Joint Space Institute Workshop 2024	
Outstanding Online Head TA of the Year, Georgia Tech	April 17, 2024
• \$500 award	
Online Head TA of the Year, School of Physics, Georgia Tech	March 7, 2024
Thomas H. Osgood Award, Department of Physics and Astronomy, MSU	April 28, 2022
• Outstanding Undergraduate Senior	
Outstanding ULA Award, Department of Physics and Astronomy, MSU	April 22, 2021; April 28, 2022
• Upper-Level Physics/Astronomy Course	
Author list, IceCube Collaboration	October 8, 2020 – Present
Dean's List, MSU	Fall 2018 – Spring 2022
Honors College, MSU	Fall 2018 – Spring 2022
Professorial Assistantship (PA), MSU	Fall 2018 – Spring 2019; Fall 2019 – Spring 2020
• Undergraduate research scholarship, ~\$8,700	

Research Experience

Wise group, Georgia Tech

August 2023 – Present

- High-Resolution Halo Resimulations

Advisors/Mentors: John Wise (Faculty)

1. Resimulating candidate halos hosting direct collapse black holes (DCBHs) at high resolution

- DCBH Classification

Advisors/Mentors: John Wise (Faculty)

1. Predicted formation of DCBHs in dark matter halos using support vector machines (SVMs)
2. Optimized SVM hyperparameters using grid-search algorithm
3. Measured importance of predictive features for classification of DCBH-hosting halos
4. Inspected halos misclassified by SVMs with different feature subsets

Li group, Georgia Tech

December 2022 – May 2023

- Black Hole Binary (BHB) Evolution

Advisors/Mentors: Gongjie Li (Faculty)

1. Simulated evolution of BHB systems around active galactic nuclei (AGNs)

IceCube Collaboration, MSU

September 2018 – Present

- Neutrinos from Dark Matter Annihilation

Advisors/Mentors: Tyce “Ty” DeYoung (Faculty), Mehr U. Nisa (Postdoc → Faculty)

1. Processing 7 years of IceCube data to use with neutrino spectra from dark matter annihilation
2. Generating custom probability distribution functions (PDFs) to calculate IceCube sensitivities to annihilation spectra
3. Tracking progress with analysis Wikipedia page and analysis GitHub repository

- Recurrent Neural Network (RNN) Event Reconstruction

Advisors/Mentors: Tyce “Ty” DeYoung (Faculty), Claudio Kopper (Faculty), Brian Clark (Postdoc)

1. Reconstructed neutrino events using RNNs for orders-of-magnitude increase in reconstruction speed
2. Gathered and processed approximately 2 million neutrino events as data for use in RNN research
3. Tracked progress with analysis GitHub repository

- Convolutional Neural Network (CNN) Event Reconstruction

Advisors/Mentors: Tyce “Ty” DeYoung (Faculty), Claudio Kopper (Faculty), Jessie Micallef (Graduate Student)

1. Optimized structure of CNNs using grid-search algorithm
2. Explored effects of 5 different loss functions on CNN regression problems for event reconstruction
3. Investigated methods for multivariate regression with CNNs

Research Supervision and Mentorship

✉ Contributed to 1 letter of recommendation

Ishita Chintala, undergraduate student, Georgia Tech

Fall 2025 – Present

- Statistical analysis of misclassifications of non-DCBH-hosting halos in an SVM
- Merger trees and growth histories of DCBH-hosting halos and misclassified non-DCBH-hosting halos

Elizabeth Mone, undergraduate student, Georgia Tech

Spring 2024 – Spring 2025

- Statistical analysis of DCBH-hosting halos
- Feature importance for classification of DCBH-hosting halos
- Decision trees for classification of DCBH-hosting halos

Publications

Published Journal Articles

2. **B. Pries** & J. H. Wise. “Identification of Candidate Halos Hosting Massive Black Hole Seeds in the Renaissance Simulations with Support Vector Machines.” *ApJ* **1004** 192, June 20, 2026 (*published online* June 12, 2026). [[ApJ](#), [arXiv](#), [ADS](#)]
1. E. Mone, **B. Pries**, J. H. Wise, & S. Ferrans. “Beyond the Goldilocks Zone: Identifying Critical Features in Massive Black Hole Formation.” *ApJ* **982** 39, March 20, 2025 (*published online* March 17, 2025). [[ApJ](#), [arXiv](#), [ADS](#)]

Manuscripts Under Review

1. R. Abbasi et al. (IceCube Collaboration, incl. **B. Pries**). “Limits on GeV-scale WIMP Annihilation in Dwarf Spheroidals with IceCube DeepCore.” *Submitted to PRD*, November 21, 2025. [[arXiv](#), [ADS](#), [InspireHEP](#)]

Dissertations and Theses

1. **B. Pries**. “Indirect Search for Dark Matter via Neutrinos from WIMP Annihilation with IceCube.” *MSU Department of Physics and Astronomy (undergraduate thesis)*, May 4, 2022.

Presentations

Invited Seminar/Colloquia/Journal Club Presentations

1. **B. Pries** & N. Willey. “Recurrent Neural Networks as a Tool for IceCube-Upgrade Reconstructions.” *Student Machine Learning Initiative, Brown University (virtual)*, October 5, 2021.

Contributed Conference Presentations

7. **B. Pries**. “Identifying Direct Collapse Black Hole Sites in Cosmological Simulations.” *The Formation and Early Evolution of Supermassive Black Holes, Baltimore, MD*, November 20, 2024.
6. **B. Pries**. “Sensitivities to WIMP Annihilation Cross Sections with IceCube DeepCore.” *American Physical Society (APS) April Meeting, Sacramento, CA (virtual)*, April 5, 2024.
5. **B. Pries**. “Sensitivities to Low-Mass WIMP Annihilation Cross Sections with IceCube Neutrinos.” *American Physical Society (APS) April Meeting, Minneapolis, MN (virtual)*, April 24, 2023.
4. **B. Pries**. “Trials Improvements for Low-Mass WIMP Annihilation Search.” *Spring 2023 IceCube Collaboration Meeting, Aachen, Germany (virtual)*, March 14, 2023.
3. **B. Pries**. “IceCube-Upgrade Reconstructions using Recurrent Neural Networks.” *2021 APS Division of Particles and Fields (APS DPF) Meeting, Florida State University (virtual)*, July 14, 2021.
2. **B. Pries**. “Update on IceCube-Upgrade Reconstructions.” *Spring 2021 IceCube Collaboration Meeting, Aachen, Germany (virtual)*, March 18, 2021.
1. **B. Pries**. “Recurrent Neural Networks for IceCube-Upgrade Reconstruction.” *Fall 2020 IceCube Collaboration Meeting, Madison, WI (virtual)*, September 14, 2020.

Contributed Conference Posters

4. **B. Pries**. “Using Neutrinos to Search for WIMPs in Dwarf Galaxies.” *University Undergraduate Research and Arts Forum (UURAF), MSU*, April 8, 2022.
3. **B. Pries**. “Searching for Dark Matter in Dwarf Galaxies Through Neutrino Production.” *Mid-Michigan Symposium for Undergraduate Research Experiences (Mid-SURE), MSU (virtual)*, July 28, 2021.
2. **B. Pries**. “Recurrent Neural Networks for IceCube-Upgrade Reconstructions.” *UURAF, MSU (virtual)*, April 15, 2021.
1. **B. Pries**. “Applications of Recurrent Neural Networks to the IceCube-Upgrade.” *Mid-SURE, MSU (virtual)*, August 10, 2020.

Contributed Seminar/Colloquia/Journal Club Presentations

2. **B. Pries**. “Limits on WIMP Annihilation Cross Sections with IceCube Neutrinos.” *Cosmic Coffee, Georgia Tech Center for Relativistic Astrophysics (CRA)*, October 11, 2023.
1. **B. Pries**. “IceCube Search for Low-Mass WIMP Annihilation in Dwarf Galaxies.” *Astronomy Seminar, MSU Department of Physics and Astronomy*, April 20, 2022.

Teaching Experience

Graduate Teaching Assistant (GTA), Georgia Tech Fall 2022 – Spring 2023; Fall 2025; Fall 2026

PHYS 6803 – Physics GTA Preparation, Steven Tarr (22 students) Fall 2026

- Conducting 2 30-minute classroom observations for each new GTA to provide feedback on teaching methods
- Assisting in in-class instruction by answering questions and guiding discussions
- Printing in-class materials

PHYS 2021 – Solar System, Dr. Paul Sell (25 students) Fall 2025

- Graded in-class worksheets and class projects
- Developed rubric for class project
- Proctored 2 midterm exams and final exam
- Held office hours twice a week to assist students with coursework by answering questions and working through problems
- Assisted in in-class instruction by answering questions and guiding problem solving
- Taught 2 class periods to fulfill Tech to Teaching certification requirements
- Graded 2 midterm exams for *PHYS 2211 – Principles of Physics I, Dr. Emily Alicea-Muñoz* (1235 students)

PHYS 2211 – Intro Physics I, Dr. Edwin “Ed” Greco (57 students) Spring 2023

Undergraduate Teaching Assistants (UTAs): Jiaying “Isobel” Deng, Amberlyn “Amber” Diehl

- Responsible for 2 lab sections with a focus on matter, interactions, and simulations
- Led lab instruction by answering questions and guiding problem solving
- Proctored and graded 3 midterm exams
- Attended weekly meetings with Head GTA to discuss lab and exam content

PHYS 2211 – Intro Physics I, Dr. Emily Alicea-Muñoz (53 students) Fall 2022

UTAs: Khushi Patel, William “Will” Wood

Head Teaching Assistant (Head TA), Georgia Tech Summer 2023 – Spring 2024

PHYS 2211 – Intro Physics I, Dr. Emily Alicea-Muñoz (1273 students) Spring 2024

24 GTAs, 31 UTAs

- Answered student questions in online forum and via email
- Held office hours once a week to assist students with coursework by answering questions and working through problems
- Wrote answer keys to exams
- Communicated with instructors regarding exam content and formatting to revise exam drafts
- Assisted GTAs in proctoring and grading exams
- Proctored and graded final exam
- Met with Course Coordinator once a week to prepare for weekly meeting with TAs
- Led weekly meetings with other GTAs and UTAs to discuss lab and exam content
- Assisted in homework debugging

PHYS 2211 – Intro Physics I, Dr. Emily Alicea-Muñoz (1137 students) Fall 2023

22 GTAs, 32 UTAs

PHYS 2211 – Intro Physics I, Dr. Andrew “Andy” Scherbakov (378 students) Summer 2023
18 GTAs

- Answered student questions in online forum and via email
- Held office hours once a week to assist students with coursework by answering questions and working through problems
- Wrote answer keys to exams
- Communicated with instructor regarding exam content and formatting to revise exam drafts
- Created exam rubrics
- Proctored 3 midterm exams and final exam for students with accommodations
- Assisted GTAs in proctoring and grading exams
- Met with instructor once a week to prepare for weekly meeting with GTAs
- Led weekly meetings with other GTAs to discuss lab and exam content

Undergraduate Learning Assistant (ULA), MSU Spring 2021; Fall 2021 – Spring 2022

AST 208 – Planets and Telescopes, Dr. Joseph “Joey” Rodriguez (39 students) Spring 2022

- Graded homework assignments and provided constructive criticism
- Held office hours once a week to assist students with coursework by answering questions and working through problems
- Assisted in astronomy lab instruction by answering questions and guiding problem solving

AST 207 – The Science of Astronomy, Dr. Gerard “Mark” Voit (66 students) Fall 2021

- Graded in-class assignments and provided constructive criticism
- Held office hours once a week to assist students with coursework by answering questions and working through problems
- Assisted in in-class instruction by answering questions and guiding problem solving
- Proctored and graded 3 midterm exams and proctored final exam

AST 208 – Planets and Telescopes, Dr. Joseph “Joey” Rodriguez (30 students) Spring 2021

Outreach, Service, and Involvement

Published Astrobites

21. **B. Pries** (ed. S. Shanbhog). “Black Holes Eating Stellar Shedding.” *Astrobites*, January 22, 2026. [[Astrobites](#)]
20. **B. Pries** (ed. C. Klare). “Connections Between X-Ray and Neutrino Emission?” *Astrobites*, December 6, 2025. [[Astrobites](#)]
19. **B. Pries** (ed. D. Solano-Oropeza). “Building Cradles for Massive Black Hole Seeds.” *Astrobites*, November 17, 2025. [[Astrobites](#)]
18. **B. Pries** (ed. W. Golay). “Stellar Feedback Regulating Massive Black Hole Growth.” *Astrobites*, November 5, 2025. [[Astrobites](#)]
17. **B. Pries** (ed. K. Lee). “Does Feedback Support Massive Black Hole Growth in Dwarf Galaxies?” *Astrobites*, October 3, 2025. [[Astrobites](#)]
16. S. Grayson, **B. Pries**, W. Smith, A. Uppal, & M. Vincent (ed. M. Ferrari & R. White). “Guide to CVs.” *Astrobites*, September 23, 2025. [[Astrobites](#)]

15. **B. Pries** (ed. K. Gary). “Eating Up the Competition: Super-Competitive Accretion in Early Star Clusters.” *Astrobites*, August 1, 2025. [[Astrobites](#)]
14. **B. Pries** (ed. C. King). “On the Straight and Narrow: How Black Hole Seeds Agree with Scaling Relations.” *Astrobites*, July 16, 2025. [[Astrobites](#)]
13. **B. Pries** (ed. S. Grayson). “Growth Spurt: Super-Eddington Accretion Leading to Overly Massive Black Holes?” *Astrobites*, July 10, 2025. [[Astrobites](#)]
12. **B. Pries** (ed. S. Roch). “Do X-Rays Help or Hurt Black Hole Formation?” *Astrobites*, May 14, 2025. [[Astrobites](#)]
11. S. Grayson, C. Meldorf, **B. Pries**, & W. Smith. “Guide to Λ CDM.” *Astrobites*, January 6, 2025. [[Astrobites](#)]
10. **B. Pries** (ed. A. Whitford). “Over the Limit: Accretion and Feedback of Early Black Holes.” *Astrobites*, December 25, 2024. [[Astrobites](#)]
9. **B. Pries** (ed. W. Golay). “Big Black Holes in Little Galaxies.” *Astrobites*, November 8, 2024. [[Astrobites](#)]
8. **B. Pries** (ed. A. Anderson). “Heavy Metal: Actinides from Compact Object Mergers.” *Astrobites*, November 2, 2024. [[Astrobites](#)]
7. **B. Pries** (ed. C. Slaughter). “Weight Gain: Growing Little Black Holes in the Early Universe.” *Astrobites*, September 24, 2024. [[Astrobites](#)]
6. **B. Pries** (ed. A. Sinha). “New Species of Dwarf Galaxy in the Galaxy Cluster Ecosystem?” *Astrobites*, September 10, 2024. [[Astrobites](#)]
5. **B. Pries** (ed. S. Grayson). “A Universal Accounting Problem: Tension in Reionization Estimates.” *Astrobites*, August 17, 2024. [[Astrobites](#)]
4. **B. Pries** (ed. V. Bonidie). “Blowout: AGNs Quenching Star Formation in Dwarf Galaxies.” *Astrobites*, June 6, 2024. [[Astrobites](#)]
3. Astrobites Collaboration (incl. **B. Pries**, ed. M. Vincent). “2024 Album of the Year: A Total Eclipse of the Sun.” *Astrobites*, April 11, 2024. [[Astrobites](#)]
2. **B. Pries** (ed. A. Masegian). “Ultra-Faint Dwarf Galaxies: Not as Small as We Thought?” *Astrobites*, April 1, 2024. [[Astrobites](#)]
1. **B. Pries** (ed. E. Clarke & N. Korhonen Cuestas). “Detecting Ghostly Neutrinos that Skim Earth’s Crust.” *Astrobites*, February 8, 2024. [[Astrobites](#)]

Astrobite Editing

24. P. Melepatte (ed. **B. Pries**). “Guest: When Neural Networks Meet Subhalos.” *Astrobites*, August 6, 2026. [[Astrobites](#)]
23. F. Morais (ed. **B. Pries**). “Guest: Trapped in the Web: The Effect of the Cosmic Web on Galaxies and Halos.” *Astrobites*, June 29, 2026. [[Astrobites](#)]
22. P. Sadhu (ed. **B. Pries**). “Guest: Novel Clues to the Formation of Ultra-Compact Dwarf Galaxies.” *Astrobites*, April 25, 2026. [[Astrobites](#)]
21. N. Kolhe (ed. C. Klare, **B. Pries**, & N. Whitsett). “Shaken and Stirred: Putting the Local Group on a weighing scale.” *Astrobites*, February 25, 2026. [[Astrobites](#)]
20. R. White (ed. **B. Pries**). “What’s the (Lagrange) point?” *Astrobites*, January 29, 2026. [[Astrobites](#)]
19. M. Verrico (ed. **B. Pries**). “Smooth like cosmic butter: How AGN destroy molecular gas clumps in galactic nuclei.” *Astrobites*, December 18, 2025. [[Astrobites](#)]

18. V. Bhandari (ed. **B. Pries**). “UR: From a Single Galaxy to the Big Picture: Using Machine Learning to Unlock JWST’s Chemical Secrets.” *Astrobites*, December 9, 2025. [[Astrobites](#)]
17. A. Gupta (ed. **B. Pries**). “How to Hide a Black Hole in a Hot Dog.” *Astrobites*, October 23, 2025. [[Astrobites](#)]
16. J. Thwaites & S. Stevenson (ed. **B. Pries** & A. Whitford). “Part 3: Extended Physics Analyses with the DESI DR2 BAO.” *Astrobites*, October 6, 2025. [[Astrobites](#)]
15. S. Stevenson & A. Whitford (ed. **B. Pries** & J. Thwaites). “Part 2: Data Analysis with the DESI DR2 BAO.” *Astrobites*, October 6, 2025. [[Astrobites](#)]
14. S. Stevenson & S. Kulkarni (ed. J. Thwaites & **B. Pries**). “Part 1: Key Results from the DESI DR2.” *Astrobites*, October 6, 2025. [[Astrobites](#)]
13. V. Bonidie (ed. **B. Pries**). “Strange New Plasma.” *Astrobites*, August 23, 2025. [[Astrobites](#)]
12. I. Oaks (ed. **B. Pries**). “Guest: What are Partial Tidal Disruption Events, and How Do We Find Them?” *Astrobites*, August 4, 2025. [[Astrobites](#)]
11. V. Cáceres (ed. **B. Pries**). “Uncovering the Precession of GW190521: How the Last Cycle Cracked the Case.” *Astrobites*, June 21, 2025. [[Astrobites](#)]
10. S. Deng (ed. **B. Pries**). “Guest: Wolf-Rayet Stars & Compact Object Collisions: A Stellar Recipe for Gravitational Waves?” *Astrobites*, May 8, 2025. [[Astrobites](#)]
9. J. Lubin (ed. **B. Pries**). “What Role Does Planet Orbital Eccentricity Play On Planet Habitability?” *Astrobites*, March 24, 2025. [[Astrobites](#)]
8. N. Sharei (ed. **B. Pries**). “Echoes of the First Light: A Mysterious Metal-Poor Galaxy at Cosmic Dawn.” *Astrobites*, March 10, 2025. [[Astrobites](#)]
7. F. Natsheh (ed. **B. Pries**). “UR: Something from Nothing: Analyzing JWST Pixel Data.” *Astrobites*, December 14, 2024. [[Astrobites](#)]
6. C. von Raesfeld (ed. **B. Pries**). “Book Review: Research is Ceremony.” *Astrobites*, November 29, 2024. [[Astrobites](#)]
5. C. Meldorf (ed. **B. Pries**). “Live Fast, Die Young, Bad Gals Do It Well.” *Astrobites*, October 23, 2024. [[Astrobites](#)]
4. S. Panjkov (ed. **B. Pries**). “Are Giant Stars Stealing Lithium from Their Neighbours?” *Astrobites*, October 17, 2024. [[Astrobites](#)]
3. N. Korhonen Cuestas (ed. **B. Pries**). “Under (Ram) Pressure! Stripping Galaxies Of Their Gas.” *Astrobites*, August 6, 2024. [[Astrobites](#)]
2. L. Rowland (ed. **B. Pries**). “The photocopied “sunburst” from the early Universe.” *Astrobites*, April 17, 2024. [[Astrobites](#)]
1. C. Slaughter (ed. **B. Pries** & K. Rockliffe). “Small but Mighty: Disk Chemistry in an M-Dwarf System.” *Astrobites*, February 19, 2024. [[Astrobites](#)]

Outreach Presentations

5. **B. Pries.** “Admitted Students Open House.” *School of Physics, Georgia Tech*, March 14, 2025.
4. **B. Pries.** “PhD Pathways Event.” *Explore Living Learning Community, Georgia Tech*, January 23, 2025.
3. **B. Pries.** “Constraining WIMP Annihilation Rates via Neutrinos.” *Astronomy Club, MSU (virtual)*, March 20, 2023.
2. **B. Pries.** “Detecting Neutrinos from WIMPs.” *Society of Physics Students (SPS), MSU*, October 21, 2021.
1. **B. Pries.** “Recurrent Neural Networks for Low-Energy Neutrino Interaction Reconstruction.” *SPS, MSU (virtual)*, October 8, 2020.

Media Appearances

2. Quoted in “IceCube sets first limits on low-mass WIMPs from dwarf galaxies.” by A. King-Klemperer. *IceCube News*, December 4, 2025. [[IceCube News](#)]
1. Quoted in “Astronomers really like the South Pole, but why?” by B. Lewis. *Popular Science*, September 5, 2024. [[PopSci](#)]

Community Service and Involvement

- Moderator/Question Judge, Georgia High School Regional Science Bowl February 1, 2025;
February 7, 2026
- Read questions for 4 matches
 - Managed buzzer system for 2 matches
- Member, Scheduling Committee, Astrobites October 23, 2025 – Present
- Wrote automatic scheduling algorithm to generate writing and editing assignments for current authors
 - Spending 3 months ensuring articles are posted every day
- Member, Editorial Committee, Astrobites August 22, 2024 – Present
- Maintaining Astrobites’ editorial standards
 - Edited 7 guest post submissions
- Member, Education Committee, Astrobites August 22, 2024 – Present
- Exploring use cases for Astrobites as an educational resource
 - Working on standardized glossary for article tags
- Moderator, Georgia High School Regional Science Bowl February 3, 2024
- Read questions for 8 matches
- Writer, Astrobites January 5, 2024 – January 26, 2026
- Summarized astronomy research papers into bite-sized articles intended for undergraduate audiences
 - Wrote 9 articles per year
 - Edited/reviewed 9 articles per year

University Service and Involvement

Georgia Tech

- Mentor, Graduate Association of Physicists (GAP) Mentorship Program Summer 2025 – Present
Mentees: Sena Ghobadi, Anne Marie Zambon (2025); Andrew Chan, Mykhailo “Mike” Yelipashev (2026)
- Meeting with 2 incoming/first-year Physics Ph.D. students approximately monthly
 - Providing advice related to classes, graduate research, finding an advisor, teaching assistantships, and life as a graduate student
- Panelist, Admitted Students Open House, School of Physics March 14, 2025
- Speaker, Ph.D. Pathways, Explore Living Learning Community January 23, 2025
- Invited panelist, Celebrating Distance Teaching and Learning Symposium October 29, 2024
- Member, GAP Fall 2022 – Present
- Panelist, Graduate School Applications panel October 12, 2022; October 30, 2024

MSU

- Mentor, Stellar Mentorship Program, Department of Physics and Astronomy Fall 2021 – Spring 2022
Mentees: Owen James, Aditya “Kal” Kalakuntla
- Met with 2 astronomy underclassmen mentees at least once monthly
 - Provided advice related to classes, undergraduate research, degree pathways, and navigating the astrophysics major
- Mentee, Stellar Mentorship Program, Department of Physics and Astronomy Fall 2021 – Spring 2022
Mentor: Jack Schulte (Graduate Student)
- Met with astronomy graduate student mentor at least once monthly
 - Discussed senior year experiences, graduate school, and graduate school applications
- Vice President, Astronomy Club Fall 2021 – Spring 2022
- Contacted astronomy faculty as potential speakers for club meetings
 - Collaborated with MSU Observatory Interim Director (Dr. Rodriguez) to plan Astronomy Club tour of the observatory facilities for approximately 20 students
 - Coordinated with President on running biweekly meetings with approximately 20 attendees
 - Helped plan and run semiannual Fall/Spring BBQ with approximately 60 attendees (joint with SPS)
- Member, Astronomy Club Fall 2018 – Spring 2022
- Member, SPS Fall 2018 – Spring 2022
- Panelist, Graduate School Applications panel March 23, 2022

Professional Societies and Affiliations

- Astrobites Collaboration January 5, 2024 – Present
- Scheduling Committee October 23, 2025 – Present
 - Editorial Committee August 22, 2024 – Present
 - Education Committee August 22, 2024 – Present
- Society for Collegiate Leadership and Advancement (SCLA) December 19, 2023 – Present

Center for Teaching and Learning (CTL), Georgia Tech	January 31, 2023 – Present
• Tech to Teaching Program	January 31, 2023 – March 10, 2026
Center for Relativistic Astrophysics (CRA), Georgia Tech	August 22, 2022 – Present
School of Physics, Georgia Tech	August 22, 2022 – Present
American Astronomical Society (AAS)	April 1, 2022 – Present
National Society of Leadership and Success (NSLS)	October 29, 2021 – Present
• Foundations of Leadership Certificate 2	April 18, 2022
• Foundations of Leadership Certificate 1	December 9, 2021
American Physical Society (APS)	May 10, 2021 – Present
Sigma Pi Sigma ($\Sigma\Pi\Sigma$) National Honor Society	April 15, 2021 – Present
Golden Key National Honor Society	November 4, 2019 – Present
Phi Sigma Theta ($\Phi\Sigma\Theta$) National Honor Society	February 18, 2019 – Present
National Society of Collegiate Scholars (NSCS)	February 10, 2019 – Present
IceCube Collaboration	August 29, 2018 – Present
Physics and Astronomy Department, MSU	August 29, 2018 – Present

Languages

- English – C2/native proficiency
- Spanish – B2/professional working proficiency
- German – A1/elementary proficiency
- Japanese – A1/elementary proficiency